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Fuzzy Logic Controller Implementation for BLDC Drive Speed Control using FOC Approach and SVPWM Based VSI Technique

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***Abstract*—Because of all of its advantages, the adoption of brushless DC motors has skyrocketed recently. It works quietly, is compact, has higher efficiency, needs less maintenance, and has many other advantages. When a motor is required to run at a certain speed, it's control is the key problem that has to be resolved. Sliding Mode Control (SMC), PID Controller, Model Predictive Control (MPC), and Adaptive Control are few of the several techniques employed. On the other hand as Fuzzy Logic Controller (FLC) is less expensive to design, is robust and doesn't require a detailed description of the system to operate; so it is used for controlling speed in this paper work. By ensuring that the magnetic fields of the stator and rotor are orthogonal to one another, a technique called field-oriented control (FOC), assures a smooth response which gives control over torque and flux. The inner loop, which measures current and outer loop which measures speed, makes up FOC's two loops. The outer speed loop is replaced from a PI controller to a FLC in the proposed work. In this article, PWM signals for a three phase inverter circuit is generated using space vector pulse width modulation (SVPWM). MATLAB and simulation were used to get the results.**

***Index Terms*—FOC, Speed Control, PI Controller, Fuzzy Logic Controller, Space Vector Pulse Width Modulation**

I. Introduction

Numerous different applications make use of brushless direct current (BLDC) motors that get their power from voltage source inverters. These motors are used in a variety of sectors due to the benefits they provide, including increased efficiency, simpler control, a longer lifespan, less operating noise, a greater torque to weight ratio, and reduced EMI interference [1]–[2].

When it comes to regulating BLDC motors, one of the approaches that is utilized most frequently is called field-oriented control. The three parameters; voltage, current, and flux may all be changed depending on whether the condition is steady-state or transient. The application of strategies allows for the distinction between torque and flux. This decoupling of the two components is referred to as "Field Oriented Control". The FOC technique involves transforming a stationary reference frame into a rotating reference frame so that torque and flux may be managed in a distinct manner. For FOC, any other control approaches, such as proportional-integral (PI) controllers, can be used to manage the torque and flux current components. These components can be controlled independently of one another [3].

Conventional controllers like PID controllers are applied to operate BLDC motors in the suitable method. However, when PID is the only controller used, parameter tweaking might be challenging. As a consequence of this, PID is used in conjunction with an advanced controller such as an adaptive controller in order to aid in reducing non-linearities. When using adaptive control, one can either directly or indirectly handle the uncertainties of the system. The presumptive parameters are employed directly in the direct technique, however in the indirect method, plant parameters are approximated and other model data is used to modify the controller. Utilizing adaptive-PID results in a combined effect that helps in responding to variations in the parameters being monitored [4].

There are several adaptive control methods that have been published in [5], [6], but only a few of them are employed in practice since they need sophisticated structures, require parameter change, and are less reliable.

[7], [8], and [9] all make use of a different controller that is known as Fuzzy Logic Control (FLC). As a result of the fact that it is widely used for non-linear systems, it is an option that is preferable when applied to non-linear systems. When compared to other controllers, FLC offers a number of advantages, one of which is the ability to automatically adapt itself for a diverse range of operational conditions. Additionally, to avoid system dispersion, a strategy based on fuzzy set theory and utilizing a trial-and-error approach is used. These fuzzy logic controllers were thus designed with non-linear systems in mind when they were initially developed.

Common PWM methods for BLDC motors include space vector pulse width modulation (SVPWM), sinusoidal pulse width modulation (SPWM), and selective harmonic elimination modulation [10]. Space vector pulse width modulation (SVPWM) is the most common kind of pulse width modulation. Compared to other PWM techniques, it offers various benefits. These benefits include a wide linear modulation range, better dc-link voltage utilization, decreased harmonic content, and flexibility in switching state selection [11].

The FOC method for regulating BLDC motor speed with the SVPWM control approach is discussed in this study. The effectiveness of PI's speed control and fuzzy logic control are then compared to one another. As proof of concept, the result obtained via the usage of MATLAB and Simulation is offered below.

II. Mathematical Modeling Of BLDC Motor

Windings on a BLDC motor includes three phases when the design of the motor is considered. Inner part is the rotor, where there are a few permanent magnets that have been placed. The modeling of brushless direct current (BLDC) motors relies on only a few assumptions which can be summarized as follows:

A. Assumptions

(a) All windings have equal resistance.
(b) Mutual and self-inductances of all windings are equal.
(c) Hysteresis and eddy current losses are neglected.
(d) Magnetic circuit saturation is not considered.

B. Nomenclature

T_e	– Electromagnetic torque (N.m)
E_a, E_b, E_c	– Back-EMFs of the motor
i_a, i_b, i_c	– Motor currents (A)
V_a, V_b, V_c	– Voltages of the phases (V)
t_a, t_b, t_c	– Torque (N.m)
θ_r	– Electrical angle (rad)
B	– Friction coefficient (Nms/rad)
J	– Moment of inertia (kg-m^2)
t_l	– Load torque (N.m)
ω_r	– Electrical speed of the motor (rad/sec)
ω	– Mechanical speed of the rotor (rad/sec)

C. Equations

Voltage equations of each phase of BLDC motor are:

$$V_a = Ri_a + L\frac{d}{dt}ia + E_a \tag{1}$$

$$V_b = Ri_b + L\frac{d}{dt}ib + E_b \tag{2}$$

$$V_c = Ri_c + L\frac{d}{dt}ic + E_c \tag{3}$$

Each winding of the BLDC motor generates back-EMF which opposes the main voltage. Equations of back-EMF are where 'L' is the inductance in Henry:

$$E_a = k_e \cdot \omega \cdot f(\theta_e) \tag{4}$$

$$E_b = k_e \cdot \omega \cdot f(\theta_e + \frac{2\pi}{3}) \tag{5}$$

$$E_c = k_e \cdot \omega \cdot f(\theta_e - \frac{2\pi}{3}) \tag{6}$$

Equations for torque can be given as:

$$t_a = k \cdot i_a \cdot f(\theta_e) \tag{7}$$

$$t_b = k \cdot i_b \cdot f(\theta_e + \frac{2\pi}{3}) \tag{8}$$

$$t_c = k \cdot i_c \cdot f(\theta_e - \frac{2\pi}{3}) \tag{9}$$

The resultant torque can be given as:

$$t_e = \frac{E_a \cdot i_a + E_b \cdot i_b + E_c \cdot i_c}{\omega} \tag{10}$$

For calculating speed we have:

$$J\frac{d\omega}{dt} + B\omega = t_e - t_l \tag{11}$$

$$\omega = \int \frac{1}{J}(t_e - t_l - B\omega) \tag{12}$$

$$\omega_r = \frac{P}{2}\omega \tag{13}$$

$$\frac{d\phi_r}{dt} = \frac{P}{2}\omega \tag{14}$$

III. SVPWM Technique

One of the most renowned technique of PWM is space vector pulse width modulation (SVPWM). More of the DC bus voltage is utilized by space vector pulse width modulation (SVPWM) than by sinusoidal pulse width modulation (SPWM). It also improves output quality while lowering harmonic distortion, which is a major plus.

The reference vector, as depicted by the three-phase system, rotates counterclockwise relative to the reference frame. The size and frequency of the fundamental component at the line side determine its amplitude and frequency, respectively. To create a reference signal from three phase references, a transformation is required. The second step, determining the reference vector, makes use of the zero vector and any neighboring switching vectors. It helps in figuring out the switching sequence of a three-phase inverter circuit [13].

The following equations may be used to calculate the voltage in this three-phase circuit:

$$V_a = V_m \cos(\omega t) \tag{15}$$

$$V_b = V_m \cos(\omega t - 120) \tag{16}$$

$$V_c = V_m \cos(\omega t + 120) \tag{17}$$

For SVPWM we first determine V_d, V_q, and $V_{\text{reference}}$.

$$|V_{\text{reference}}| = \sqrt{V_d^2 + V_q^2} \tag{18}$$

where,

$$V_d = V_{an} - \frac{1}{2}V_{bn} - \frac{1}{2}V_{cn} \tag{19}$$

$$V_q = V_{an} + \frac{\sqrt{3}}{2}V_{bn} - \frac{\sqrt{3}}{2}V_{cn} \tag{20}$$

Now $\alpha = \omega t = 2\pi f t$

The time duration T_1, T_2, and T_0 calculated using following equations:

$$T_1 = \frac{t}{2\sqrt{3}} \frac{V_{\text{reference}}}{V_1} \sin\left(60 - \phi\right) \tag{21}$$

$$T_2 = \frac{t}{2\sqrt{3}} \frac{V_{\text{reference}}}{V_2} \sin(\phi) \tag{22}$$

$$T_0 = t - (T_1 + T_2) \tag{23}$$

where $t = \frac{1}{f}$, with f being the frequency at switching mode, and ϕ is the angle between each vector, ranging from 0 to 60 degrees.

IV. Field-Oriented Control Based PI

The Field Oriented Control (FOC) technique is broken down into its component parts in the figure (1) that may be found below. The management of stator currents is made possible through the use of a control method known as field-oriented control. Because of the decoupling that takes place as a consequence of this, torque and flux may now be managed independently. The phase currents of the stationary reference frame are converted into the torque and flux producing current components of the rotating reference frame through this method. These currents are I_d and I_q. The information from the stationary reference frame must be transmitted to the rotating rotor reference frame in order to accomplish dq. We employ the Clarke and Park transformations to do this.

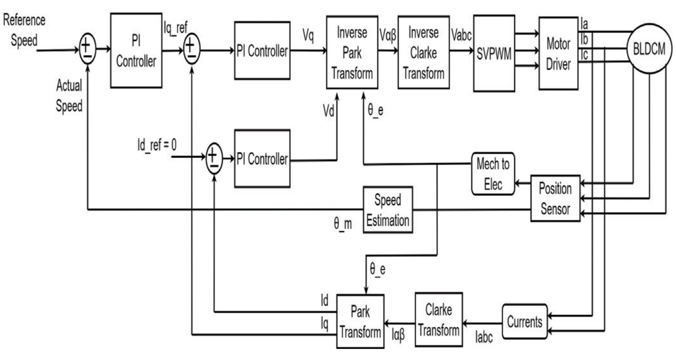

Fig. 1. Block Diagram Of Speed PI Based FOC

The block diagram depicts a vector-controlled technique where the inner loop is controlled by two PI controllers and the outer loop by one PI controller. The PI controller of the outer loop, which serves as a reference input for the torque component ($I_q ref$), computes error based on the reference speed and the actual speed provided to it. Reference flux component ($I_d ref$) is kept to zero in order to maximize desired useful torque. Then both current components (I_q, I_d) are compared to the corresponding reference values and passed on to PI controllers which gives $V_{d_{\text{ref}}}$ and $V_{q_{\text{ref}}}$. These voltage values in the frame known as rotating reference are transformed using the reverse Park's transformation to a two-axis stationary reference frame to obtain values V_α and V_β. Then converted to V_{abc} using inverse Clarke's transformation which act as inputs for SVPWM block.

V. FOC Based Fuzzy Logic Implementation

A. Block Diagram of Fuzzy Logic Control

An approach to computing known as fuzzy logic uses degrees of truth rather than the traditional binary logic of either 1 or 0. One of the numerous applications for this control mechanism is the regulation of speed. When an application such as speed control is taken into consideration, fuzzy logic is used to adjust the speed of the motor in response to changing parameters such as load, temperature, and other such elements. FLC use a strategy that is distinct from traditional controllers whenever they are tasked with decision making. They do not necessitate the use of mathematical models with a high level of detail. These controllers are reliable for use in non-linear systems due to how effectively they manage irregular conditions and faulty data. The usage of these controllers in non-linear systems is recommended.

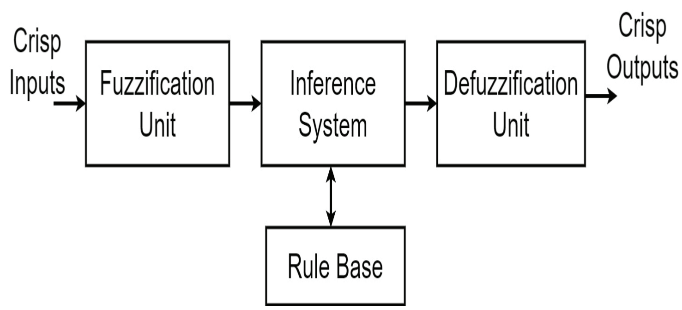

Fig. 2. Block Diagram for FLC

An example of a diagrammatic representation of fuzzy logic control may be seen in the following image (Fig. 2). As a direct consequence of this, the fuzzy logic system equips each block with specific responsibilities. Crisp values are turned into fuzzy values through a fuzzification unit. The rule base unit is responsible for the storage of process information. The fuzzy rules are applied to the rule base by the inference block, which then produces fuzzy output with a value that is somewhere between 0 and 1. Therefore, a defuzzification unit is necessary in order to utilise this output for the system. This unit turns fuzzy values back to crisp values [12]. As can be seen in the picture below, the block diagram for the fuzzy-interfaced FOC looks like this. The speed PI controller has been replaced with fuzzy logic, and three-phase inverter circuitry is being used to feed it to a BLDC motor. The BLDC motor employs the SVPWM approach, and fuzzy logic is in charge of controlling it.

B. Fuzzy Logic Implementation

Fig. 3 shows FOC using FLC as speed controller. This example uses the Mamdani method with the triangle function. The seven membership functions that are used are Negative Small (NS), Negative Medium (NM), Negative Big (NB), Zero (ZE), Positive Big (PB), Positive Medium (PM), and Positive

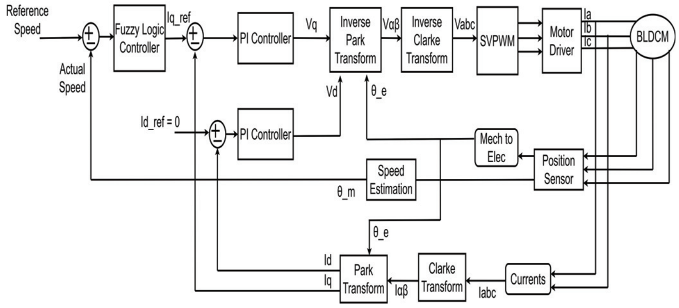

Fig. 3. Block Diagram for FLC

Small (PS). The two inputs that are taken into account by this system are error and the change in error [12]. The output value is the reference current value I_qref. The result will be more accurate when more membership functions are used. Its development incorporated the center of gravity (COG) approach of defuzzification [12]. The reality of this is seen in the table (I) below:

TABLE I
RULE-BASE FOR FLC

e/ce	**NB**	**NM**	**NS**	**ZE**	**PS**	**PM**	**PB**
NB	NB	NB	NB	NB	NM	NS	ZE
NM	NB	NB	NB	NM	NS	ZE	PS
NS	NB	NB	NM	NS	ZE	PS	PM
ZE	NB	NM	NS	ZE	PS	PM	PB
PS	NM	NS	ZE	PS	PM	PB	PB
PM	NS	ZE	PS	PM	PB	PB	PB
PB	ZE	PS	PM	PB	PB	PB	PB

VI. SIMULATION AND OUTPUT

A simulation model was made in MATLAB/Simulink, as can be seen in figures 1 and 3, with the goal of evaluating the approach for the BLDC drive that was previously advised for the BLDC drive. The first model is a BLDC drive that utilizes FOC and speed PI as its primary calculus. After that, a comparison is carried out by substituting a fuzzy logic controller for the speed PI method. The same data from the simulation was used. The speed controller's principal objective is to maintain the reference speed that it was set at. In the next table (Table II), data of the motor parameters that were utilized for the simulation is showed.

TABLE II
MOTOR PARAMETERS

Parameter	**Value**
DC Input Voltage	24 V
Resistance of the stator per phase	0.05 Ω
Inductance of the stator per phase	0.56e-3 H
Rated Torque	0.06 N.m
Moment of Inertia	0.027 kg
Pole pairs	6"

Following are the output graphs (Fig. 4 to Fig. 9) obtained for the above mentioned methods.

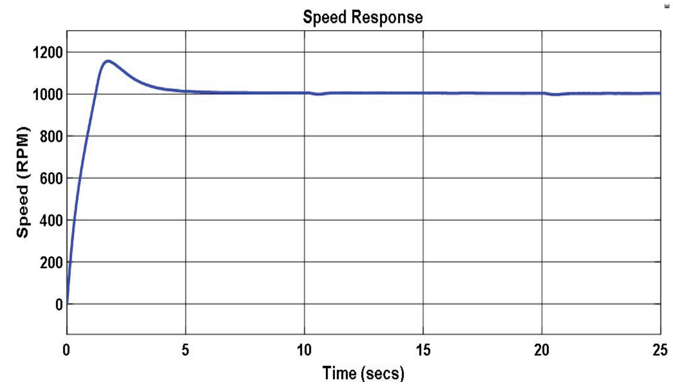

Fig. 4. Speed Response for PI Controller in RPM

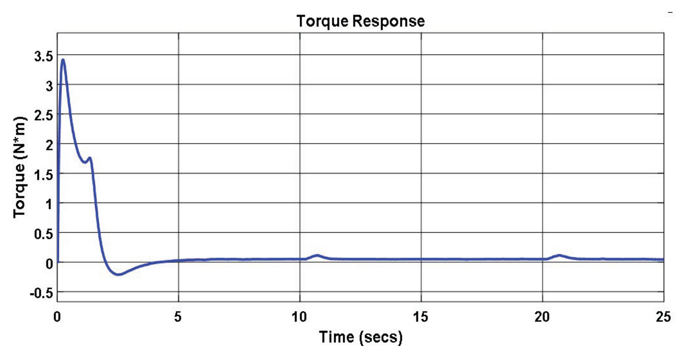

Fig. 5. Torque Response for PI Controller in Nm

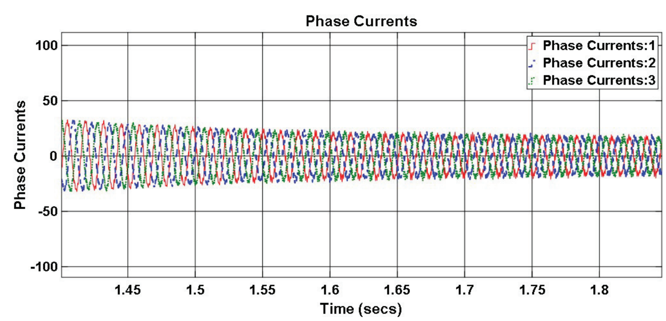

Fig. 6. Phase Currents Response for PI Controller

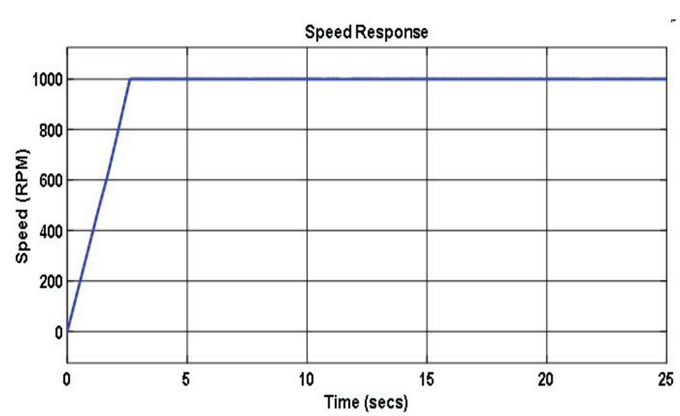

Fig. 7. Speed Response for FLC Controller in RPM

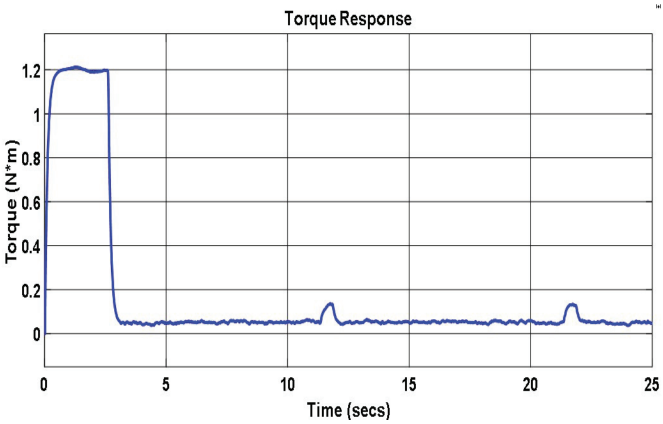

Fig. 8. Torque Response for FLC Controller in Nm

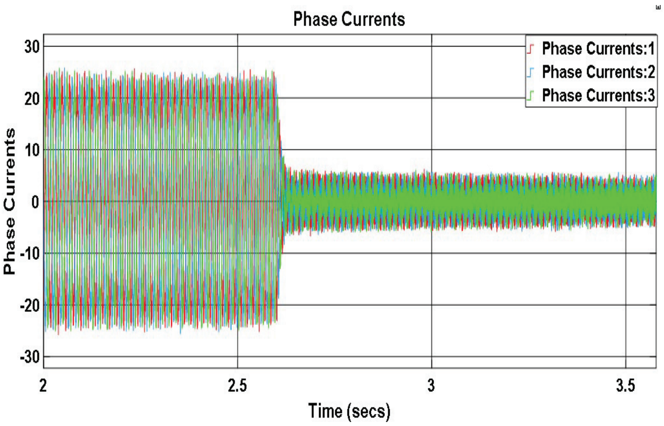

Fig. 9. Phase Currents Response for FLC Controller

VII. Results and Discussion

Regarding the previously stated field-oriented control strategy, which employs PI and FLC as its speed loop controller, the following findings were observed. FLC provides a better response than PI controller because the PI controller's speed response curve takes a longer settling time—22 seconds—to achieve steady state, compared to FLC's three-second settling time. The PI graph shows one overshoot with a settling period of about 7 seconds when speed graphs are taken into account. In contrast, FLC does not overrun and has a direct transition at 2.60 seconds. The torque response exhibited two overshoots and one undershoot when utilizing a PI controller, and the settling period was around 7 seconds. The highest torque, however, was 1.3 N.m when utilizing an FLC controller, and the settling time was around 2.80 seconds at an interval of 0.08 N.m. One can observe multiple little peaks after a constant period of 14 seconds. The FLC's settling time is two minutes and 52 seconds when the phase currents condition is taken into account, whereas the PI controller's settling time is two seconds for the same.

VIII. Conclusion

In this article, equations for mathematical modeling BLDC motors are created. In addition, Field Oriented Control (FOC), which makes use of a PI and FLC controller, is utilized in order to accomplish the goal of having full control over torque and flux. SVPWM is a well-liked PWM approach for three-phase inverter circuits because it offers optimal output, decreased commutation losses, and harmonic loss reduction. These characteristics make it an attractive choice. The creation of equations for controllers is based on fuzzy logic. To do this, MATLAB and Simulation were utilized throughout the process. Both the PI controller and the FLC simulations are demonstrated, and their differences were examined. The results show that FLC has a greater level of accuracy and a quicker speed response when compared to PI controller.

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